David Solano

Calgary, Alberta | david.solano0813@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Early-career software, automation, and systems-integration professional with a completed Bachelor of Science and professional testing responsibilities since 2021. Independently designed and built three end-to-end applications spanning Python/PyQt6 with Qt/QML embedded-device verification, field communication surveys with geospatial visualization, and Linux/Playwright workflow automation. Academic geospatial experience includes spatial-data concepts, geospatial communication, visualization, and ArcGIS. Experienced troubleshooting across software, firmware, hardware, configuration, connectivity, and installation while supporting deployed industrial communication systems and collaborating with firmware, hardware, field-support, and software teams.

TECHNICAL SKILLS

Programming & Data:	Python, C, C++, Java, JavaScript, SQL, MicroPython; database synchronization, data processing, CSV reporting
Software & QA:	Application design and development; functional, system, regression, release, and supplemental exploratory testing; test automation; defect investigation; root-cause troubleshooting
Systems & Interfaces:	Windows, Linux command line, Raspberry Pi, extensive SSH use, Qt/QML, embedded systems, firmware testing, serial/UART, OpenThread, Ethernet, Wi-Fi, LTE/cellular, RS232/RS485, USB, GPIO, SD cards, sensors
Geospatial & Visualization:	ArcGIS (academic), geospatial data and communication, spatial visualization, field survey data, Google Earth-compatible output, CSV reporting
Tools & Delivery:	Azure DevOps repositories, commits, branches, pull requests, work items, defects, test cases, and pipelines; Git; Playwright; GitHub/GitLab and Jira in university-project contexts

PROFESSIONAL EXPERIENCE

BossPac Engineering & Technology

October 2019 - Present
Professional testing responsibilities since 2021

Computer Science Student

Calgary, Alberta

- Independently designed and built the BossPac Quality-Control Testing Application from scratch using Python/PyQt6 with Qt/QML for Windows, automating embedded-device commands, serial/UART communication, expected-response validation, timed echo testing, configuration checks, and repeatable setup and cleanup.
- Developed and executed functional and system tests through repeatable protocols and QA checklists aligned with Product Verification Test documentation; retested new and changed functionality before releases and produced version-specific coverage documentation for firmware and hardware teams.
- Validated DNET, DNET+LTE, and satellite communication products across Ethernet, Wi-Fi, LTE/cellular, RS232, RS485, USB, GPIO, SD cards, sensors, power-loss and reboot-recovery scenarios, and intermittent-connectivity conditions.
- Created structured Azure DevOps defects containing reproduction steps, expected and actual results, and logs or other supporting evidence; investigated failures across software, firmware, hardware, configuration, installation, and connectivity.
- Used Azure DevOps professionally for repositories, commits, branches, pull requests, work items, defects, test cases, and pipelines to manage application changes, testing evidence, and cross-functional follow-up.
- Supported deployed industrial communication systems through field troubleshooting, installation and maintenance guidance, configuration review, connectivity diagnostics, and verification that documented procedures were completed correctly.
- Worked directly with firmware and hardware personnel to clarify expected behaviour, outputs, device targets, and additional coverage; contributed technical procedures, troubleshooting guidance, training materials, and Product Verification Test records.
- Onboarded and trained a new team member in testing, troubleshooting, technical-problem identification, defect tracking, documentation, and review of testing work and findings.

Vans

July 2017 - June 2019

Sales Associate

Calgary, Alberta

- Delivered customer service, product guidance, sales support, and issue resolution in a fast-paced retail environment while communicating with customers from varied backgrounds.
- Collaborated with team members to maintain floor operations, product organization, inventory awareness, and a consistent customer experience during high-volume periods.

SELECTED TECHNICAL PROJECTS

Bifrost Automation System Independent Build — September 2025 - Present

- Independently conceived, designed, and built an end-to-end Linux/Raspberry Pi automation system for member selection, room booking, email verification, confirmation processing, scheduling, reporting, and database synchronization.
- Developed Playwright browser workflows with business-rule and input validation, stage filtering, retry logic, failure recovery, and multiprocessing; deliberately compared front-end workflow behaviour with backend/database state.
- Implemented and debugged email scanning and confirmation processing, documented and investigated state discrepancies, generated CSV reports and daily summaries, and deployed the system using Python, Playwright, Linux, Raspberry Pi, and SQL.
- Translated firsthand ownership of a recurring student-organization scholarship process into documented requirements, user support, workflow validation, and continuous improvement.

Radio Survey Tool Independent Build | BossPac Internal Tool — April 2024 - Present

- Independently designed and built a Python tool that transformed Radio/DNET field-survey data into processed results, CSV reports, and Google Earth-compatible geospatial visualizations for wireless-coverage analysis.
- Personally conducted field trials, evaluated communication reliability and device connectivity, processed collected spatial data, and used the results to support installation decisions and field troubleshooting.

University of Calgary Solar Car Dashboard System Collaborative Team Contribution — September 2024 - April 2026

- Contributed to Qt/QML real-time dashboard interfaces displaying vehicle telemetry and embedded-system data; collaborated with embedded, electrical, telemetry, and software teammates.
- Supported hardware-software integration and troubleshooting during system testing and competitive racing events, using Jira in the university-project environment to coordinate technical work.

ADDITIONAL TECHNICAL & LEADERSHIP EVIDENCE

- Reviewed automated-test code written by another person and analyzed an existing codebase to identify where automated tests should be introduced; kept these activities separate from unverified project attribution.
- Performed supplemental randomized exploratory testing after documented protocols passed to identify unexpected behaviour outside standard procedures.
- Helped onboard and coach university students through code improvements, problem-solving approaches, technical analysis, and documentation guidance.
- Completed GEOG 280 (Thinking Spatially in a Digital World) and GEOG 380 (Geospatial Communication), including academic work with geospatial data, visualization, and ArcGIS.

EDUCATION

Bachelor of Science in Natural Sciences — **University of Calgary**

Primary concentration: Computer Science | Secondary concentration: Mathematics — Degree requirements completed

Convocation November 9, 2026

Relevant coursework: Introduction to Software Engineering; Database Management Systems; Computing Machinery II; Data Structures, Algorithms and Analysis; Software Analysis; Explorations in Software; Thinking Spatially in a Digital World; Geospatial Communication.

ADDITIONAL INFORMATION

Language:	Spanish - native speaking and writing	**Work status:**	Canadian citizen
Mobility:	Open to remote, hybrid, on-site, and relocation opportunities	**Licence:**	Valid Alberta Class 5

REFERENCES

Anthony Bastiaansen	Engineer, BossPac Engineering & Technology	403-809-9088
Darren Sortland	President, BossPac Engineering & Technology	403-796-3975
Kyle Mulligan	Assistant Vice President - Operations Technology, CPKC	587-700-1112
William Bailey	Petroleum Oil & Gas Engineer, Corncracker	403-923-8638